

Homework 6

COMP221 Spring 2026 - Suhas Arehalli

Complete the problem below. Check the course website & syllabus for further instructions.

If any problem is unclear, or you think you found a typo, please let me know ASAP so I can clarify!

When showing the correctness of an algorithm (including for reductions), a brief sketch of the proof will suffice. Your proposed algorithms must still be correct to receive credit, and sketches should be clear and precise enough to allow for a peer to reconstruct a more detailed argument without any new insights into the problem.

Problems

1. Degree-Constrained Spanning Trees (*Based on Skiena 11-12*)

- (a) The low-degree spanning tree problem (LOW-DEGREE-SPAN) is a decision problem defined as follows:

PROBLEM (LOW-DEGREE-SPAN).

Input: An undirected graph $G = (V, E)$ and $k \in \mathbb{Z}_{\geq 0}$

Output: TRUE iff there exists a spanning tree where each vertex in the tree has degree $\leq k$. FALSE otherwise.

Prove (using a reduction) that LOW-DEGREE-SPAN is NP-hard. Make sure you provide both a description of the reduction and a brief justification of the correctness of the reduction as part of your answer.

****Hint**:** Skim through the NP-hard problems you've seen so far (including those in the textbook we didn't discuss in class!). Which involve trees with constrained degree? This might require some clever thinking!

- (b) Consider the *high-degree spanning tree problem* (HIGH-DEGREE-SPAN) which is defined as follows:

PROBLEM (HIGH-DEGREE-SPAN).

Input: An undirected graph $G = (V, E)$ and $k \in \mathbb{Z}_{\geq 0}$.

Output: TRUE if G contains a spanning tree which contains a vertex with degree $\geq k$ (within the spanning tree). FALSE otherwise.

Show that HIGH-DEGREE SPAN is in P.

****Hint**:** Think through what graph methods we've seen that create spanning trees. Can one force a high degree vertex (and its edges!) into a spanning tree?